

expectation of the user. Depending on that, you will have to tweak your algorithms and data flow.

The applications aka use-cases

From the consumer side, there are many applications of doing AI or ML on edge—from voice assistant, face recognition to wearables. From a B2B perspective, it is Industrial IoT (IIoT).

“A French company, which manufactures components for aircraft, uses real-time location solutions (RTLS) like RFID or BLE to track different parts. Since new parts keep coming in when old parts are consumed, the company was not able to understand the usage frequency of a particular component in a manufacturing line. Therefore they made gateways, which were deployed in the warehouse and started implementing ML on them. The gateway was power over ethernet (PoE), which has a good processor capability. Through this, they were able to process the information locally and the operator on the floor got quickly informed about the outcome. This was done through multiple ML algorithms on data over a period of one year. This was not done real-time but based on predictions made by the algorithm. Ninety per cent of the time, the predictions came out to be true. If it was done real-time, it would have taken a lot of time to make a decision and give the output to the operator,” informs Vinay Solanki.

He further adds that his company, Napino, leverages AI on edge as robots are used for evaluation purposes. An in-house solution is also being built for social distancing and marking attendance of employees using IoT applications with beacons. This is not yet an AI implementation, but the application is going to have a lot of data on which ML will be implemented to find analytical insights. Another application that was worked upon was a smart

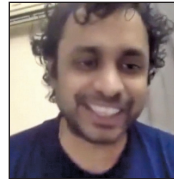
Speakers

Amit Kurhekar,
head of Data
Science, Yodlee



Dharmendra Kumar,
head of IoT,
Arihant

Nikhil Bhaskaran,
founder,
Shunya OS



Rishi Gargi,
founder,
Thingify

Venkateswar Melachervu,
CTO, Brillium
Technologies



Vinay Solanki,
head of IoT,
Napino

then data acquisition will be fast and happen inside edge only rather than data going to the cloud and then giving output. On the consumer side, home automation seems to be big.

Venkateswar Melachervu states a very unique use-case. A year or two ago, he was working on providing a telemedicine platform for pregnant women—right from conception to postpartum. Since a lot of doctor consultations and tests take place over a period of nine months, the idea was to offer these services at convenience with the help of an app. This was a noble step in timely addressing certain critical diseases that can manifest during the pregnancy such as gestational diabetes mellitus, which is characterised by sugar levels becoming high during pregnancy, affecting the health of both the baby and the mother.

Because of the computational intensity and data, it was chosen to run a model in the cloud, get the data and give the results. But with AI-edge coming in, a model can now be created on the edge itself, based on the data of 10,000 to 20,000 pregnant women. With sufficient data, it is possible to derive a linear prediction model to arrive at a risk percentage (between 0 and 1). It can help in predicting early-on gestational diabetes with the help of ML, which can be constantly updated as the results keep coming in.

He also states another use-case. Electricity meter reading is generally not done manually now. At times, this leads to inaccurate billing. With a little bit of built-in edge intelligence on a handheld device, a meter reader can go to homes and use that device to generate bills right then and there. This will make it possible to accurately measure the consumption units in a single day without any error, cutting down lots of problems for the consumers as well as the electricity board.

shower, which monitored the amount of water consumed while taking a bath. That was achieved using a flow meter plus an IoT gateway to upload the data.

According to Dharmendra Kumar, a major application for AI on edge is industry asset monitoring. Machines need to be controlled and monitored for optimal data consumption and processing. If this happens on AI-edge,